

Natural dyes and pigments in functional finishing

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1 Introductions

Textile is an essential and vital aspect of human existence, and it also plays an important role in meeting human requirements. They endow in giving garments and attires used by mankind, as well as determining their lifestyle. Initially, garments or textiles were considered to just cover the human body while also providing protection from the surrounding environment, but as time progressed, numerous types of textiles came into being that are now used in daily human activities. Textiles are woven and processed materials formed from yarn and fibers taken both from natural and man-made materials. The growth of the textile industry has recently been greatly aided by the use of numerous technical innovations and scientific advances in the manufacture of cotton, silk, and wool [1]. The net value of the textile industry in the present situation, encompassing textile and fabric production, manufacturing, and sales, was projected to be three trillion US dollars. The textile business has a broad and diverse range of products with an even wider range of applications. The material sector is one of the most important for the financial success of people throughout the globe due to its wide diversity, which is only one of many contributing factors [2].

Processes including coloring, preparation, and functionalization—which are sometimes expensive, energy-intensive, and chemically hazardous—are included in the dyeing and finishing sector. In the dyeing and finishing industries, some qualities including color, shade, pattern, quality, and kind of processing are crucial for boosting the fabric's attributes [3]. Currently, throughout their processing stage, these wet chemical processing machines use a lot of water, colors, chemicals, and accessories. Additionally, this industry generates a huge amount of industrial effluent that is discharged into the neighboring aquatic streams, causing ecosystem degradation and a lack of drinking water. Researchers have made various efforts to include sustainability in the dyeing and finishing processes because of the negative impacts of textile effluent, such as water pollution, climate change, energy consumption, and water shortage [4]. To lessen the negative impacts of synthetic colors, researchers have looked at natural dyes made from plants such as turmeric, tulsi, lemon, lavender, and so on. In the finishing process, a variety of ingredients are utilized to improve the strength, stability, wear resistance, water resistance, flame resistance, and UV protection. These materials include plant extract, biopolymers, biomolecules, proteins, agricultural derivatives, and lignocellulosic compounds [5].

Finishing textiles is a crucial step in raising the value of finished textile goods, and many finishing procedures include hazardous chemicals and other materials. Due to the increased public awareness of environmental issues, producers and consumers alike now demand sustainable solutions across the whole supply chain [2]. It goes without

saying that textile producers are also looking for environmentally friendly finishing methods and procedures that use natural and secure components to offer value. Advanced finishing techniques including encapsulation techniques and plasma treatments raise the affinity and durability of the processes, which increases sustainability. Natural colorants may now be used to impart a variety of functional finishes due to the value-added textiles and clothing industry's fast rise. Due to customer demands for a healthy, hygienic, and pleasant lifestyle, textile substrates with ultraviolet protection, mosquito-repellent, insect-repellent, and antioxidant textiles play an important role in the global market [6]. The use of natural dyes to provide UV-protective, mosquito-repellent, insect-repellent, and antioxidant textile qualities has been researched and documented.

Therefore, this chapter discusses the various functional finishing on textile materials using different types of natural dyes and pigments. The application areas of various functional finishing including UV-protective, mosquito-repellent, antiodor, insect-repellent, and antioxidant textiles are well described. The different natural dyes and pigments used for the functional finishing are included with descriptions and chemical structures.

2 Functional finishing

Functional finishing is done in the textile material to impart different characteristics such as durability, comfort, and repellency from different environmental factors. Different types of agents, methods, and procedures are applied to achieve the best performance in this regard. The classification of functional finishing can be illustrated in the following Fig. 1.

2.1 UV-protective finishing

The UV radiation from the sun, having a larger wavelength than visible light, can lead to serious damage to textile materials or human health, especially on the skin by the occurrences of some chemical reactions. To protect the wearers' skin from these harmful effects of sun radiation, UV-protective finishing agents, also known as UV shielding agents can be applied to textile materials. A little exposure to the sun may be beneficial to the health or may not cause any serious damage to the textile. However, overexposure to the sun leads to harmful effects including skin pigmentation, sunburn effects, aging, skin cancer, and even DNA damage as various cellular responses are induced by UVA (320–400nm) and UVB (280–320nm) rays. In addition, UV radiation causes photochemical degradation, fading, increased crystallinity, and other physical and chemical changes [7]. As a result, UV protection finishes are essential to apply in a wide range of textile applications (Fig. 2A) such as activewear, high-altitude garments, wearable devices, covering materials, and other technical textiles.

Classification	Finishes
Stabilization Finish	➤ Durabe Press Finish
Appearance Retention Finish	➤ Soil Release Finish. ➤ Abrasion Resistace Finish. ➤ Crease-resistance Finish.
Comfort-related Finish	➤ Water proof Finish ➤ Softenig Finish ➤ Hydrophilic Finish
Biological-control Finish	➤ Antimicrobial Finish ➤ Anti-odor Finish ➤ Antioxidant Finish
Safety-related Finish	➤ Flame Retardant Finish ➤ Water-repellent Finish ➤ UV-protection Finish

FIG. 1 Classification of different functional finishes.

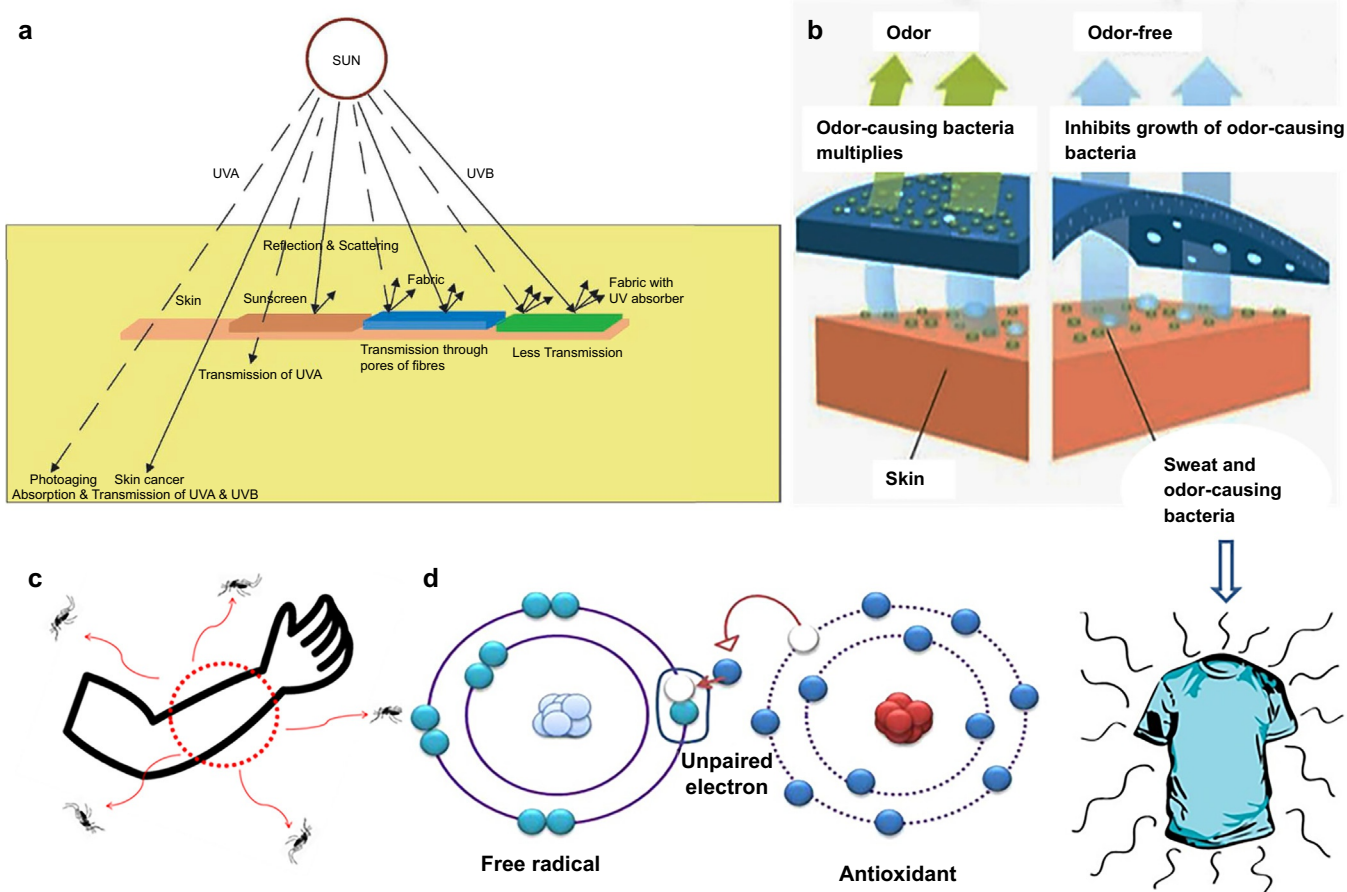


FIG. 2 Different types of finishing agents: (A) Protection of skin by UV-repellent protective agents [8]. (B) Illustration of creating odor in clothes. (C) Simple representation of repelling mosquitoes and insects. (D) Illustration of a chemical reaction between free radicals and antioxidants. Panels (A and B) Reproduced from Ref. [9], Copyright CC-BY terms, 2020. Panels (C and D) Reproduced with permission from Ref. [10], Copyright 2021 Springer.

2.2 Antiodor finishing

Odor creation is very common for clothing and it also determines the comfort, cleanliness, and sanitation of any fabric. To protect the fabric or clothing from any fungi, bacteria, or any microorganism that are the main reasons of odor creation and odor retention, the application of antiodor agents is very common in the textile sector, especially focusing on some functional clothing such as sportswear and activewear that face the problems of sweating. Sweating is a very basic biological mechanism where sweat glands produce a mixture of salt, water, and other constituents subsequently cooling off the body by evaporation technique. After secretion of the sweat on the body's surface, the microorganisms such as bacteria consume the sweat as their nutrition source by chemical synthesis of sweat that leads to odor creation (Fig. 2B) [11]. Another common place of odor creation is socks where bacteria get favorable dampness, warmth, and dark surroundings to grow easily. Protecting textile components from odor creation often becomes complex because the creation of odor varies from person to person according to their food habits, smoking, and other life-style things [11]. Though antimicrobial finishes are often used as a complementary material, there are a few limitations that remain in this regard such as the antimicrobial finishes are not much durable and there is much possibility of them going away after several washings [11]. The available antiodor agents also have some limitations. For example, some antiodor agents interact with the skin creating skin irritation and some of them are susceptible to carcinogenicity [12]. Therefore, the investigation regarding the production of antiodor agents should be precise so that proper laundering of the clothing is possible and the agents do not create any harmful effects because of human skin exposure. A recent investigation has been done to produce microencapsulated agents for the slow release of the fragrance that can confirm social comfort and esthetic appearances [13].

2.3 Mosquito-repellent finishing

The application of the mosquito-repellent agent in textile materials is one of the most growing ways in the advancement of clothing technology, especially where the dissemination of some detrimental diseases such as dengue, malaria,

filariasis, and chikungunya is very high [14]. As mosquito is very well known as a vector with deadly nature to spread various diseases around the places, various mosquito-repellent products like lotion, coil, and liquids are common in the market. However, some limitations such as breathing problems, choking issues, and skin irritation of these products lead to the application of mosquito-repellent agents in the fabric and clothing materials that are much capable to keep the mosquito away (Fig. 2C) [15–18]. The action of mosquito-repellent agents can be broadly divided into two types: 1. olfactory mode, and 2. tactile mode. In olfactory mode, also known as transpiration repellency, the humidity sensing mechanism is disabled which leads to distracted mosquitoes to locate humans. On the other hand, the tactile mode affects the mosquito's nervous system keeping them in a confused state and resisting them to bite the target [19]. Mosquito-repellents can be classified on a different basis—chemical or herbal repellents depending on origin. They can also be classified based on their action. Repellent insecticides repel insects and pests rather than killing and contact insecticides disturb the nervous system when coming into their contact.

2.4 Insect-repellent finishing

Besides mosquitoes, some other insects act as vectors to carry different types of diseases. For example, ticks are also found very harmful causing various fevers including Lyme diseases and hemorrhagic fever. Plague is spread to humans initially carried out by rodents, rats, and dogs and then transferred to the human body [20]. Ant is another very common insect found in maximum regions, especially with the largest population in rainforests. The effect of ant bites can be from slight pain to serious detrimental issues. For example, fire ant stings can cause white pustule formation, pain in the chest, dizziness, nausea, and in some cases, serious shock or coma. In some regions of the earth, chiggers are commonly found and they cause scrub typhus or tsutsugamushi disease. Several options for products in the market can protect the human body safely to a large extent. However, insect-repellent textile materials or clothing are found most effective for the protection of the wearer. The classifications of insect-repellent textiles are insecticidal or repellent. The insecticidal are found for application in the treatment of home textile applications. The applications of insect-repellent agents are found in the factory. The major concerns regarding the application of insect-repellent agents are the longevity of the agents and the safety issues when chemical agents are used. Consequently, sustainable and natural sources can be the alternative in this regard.

2.5 Antioxidant finishing

Antioxidant contains some molecules that can prevent the oxidation process of other molecules. In general, oxidation is a chemical reaction that can damage any material or human skin by a chain reaction followed by the production of reactive oxygen species (ROS), or free radicals [10]. A free radical is known as one unpaired electron that is typically released during metabolism [21]. Apart from this, different environmental factors such as polluted air, smoked air, alcohol, toxins, etc., may also cause the creation of free radicals that are threatening the health and hygiene of human life. Free radicals including hydroxyl radicals ($\text{OH}^\cdot$), superoxide anion radicals ($\text{O}_2^\cdot$), and especially nonradical species like H_2O_2 and singlet oxygen are some of the components of ROS, which is a kind of activated oxygen ($^1\text{O}_2$). ROS is the primary cause of oxidative stress. If the textiles are treated with antioxidants, it gains the ability to be a medium for delivering antioxidants to the skin. As a result, the free radicals caused by skin degeneration can be deactivated leading to skin protection from oxidative stress and damage [22,23]. The mechanism is described in Fig. 2D. There are numerous antioxidants in nature such as *Aloe vera* extract that neutralize the effects of ROS. Phenol compounds can trap free radicals, and scavenge them through a series of linked processes in the presence of antioxidant enzymes, or remove them directly [24]. Consequently, the availability of the integral phenolic OH in *A. vera* extract can work as an antioxidant agent [25].

3 Testing procedure for functional finishing

3.1 UV-protective finishing

3.1.1 Test procedure

There are several techniques for evaluating UV-protective finishing when applied to textile materials. Based on various factors such as UV radiation source, geography, level of coverage of body parts, and the time of exposure to the sun, the selection of standards and methods vary in different countries. The incorporation of information about UV protection capacity along with other parameters of any clothing can help the consumers be aware while using.

A variety of techniques have been used to evaluate the fabric's suitability for ultraviolet protection. In vitro testing and in vivo testing are the two main types of quantitative testing.

In vitro testing

A spectrophotometer or spectroradiometer is used to determine the UPF of the fabrics using the in vitro photoprotective method [26,27]. It describes how much effective any fabric can be against the harmful ray of the sun. A higher UPF number denotes greater fabric protection levels. To calculate the fabric's spectral transmittance, wavelengths between 280 and 400 nm have traditionally been utilized. UPF is defined as the difference between ED and ED_f, which represent the UV radiation irradiance computed via the fabric and the UV radiation irradiance transferred through the air, respectively. To calculate the UPF value, the following Eq. (1) can be used.

$$\text{UPF} = \frac{\text{ED}}{\text{ED}_f} = \frac{\sum_{\lambda=280}^{\lambda=400} E(\lambda) S(\lambda) \Delta \lambda}{E(\lambda) T(\lambda) S(\lambda) \Delta \lambda} \quad (1)$$

Here,

$E(\lambda)$ = relative efficacy of the erythema spectrum (unitless).

$S(\lambda)$ = solar ultraviolet radiation (UVR) spectral irradiance in $\text{Wm}^{-2} \text{nm}^{-1}$.

$T(\lambda)$ = average fabric swatch spectrum transmission.

$\Delta \lambda$ = wavelength interval in nm.

UPF calculation is a universal term for different countries around the world to evaluate, categorize, and characterize a fabric's UV protection capacity.

In vivo testing

Sun protection factor (SPF) measurements can be made in vivo, commonly known as on human skin or in a lab (SPF). SPF is a word that has mostly been associated with sunscreens. SPF is a comparison of the UV exposure on skin that is protected (MEDps) to skin that is exposed (MEDus) in joules. A higher level of SPF implies better protection against sunburn [26].

3.1.2 Standards and methods of analysis

Global solar UV index (UVI)

The official International Commission on Illumination (CIE) standard action spectrum for UV-induced erythema on human skin described the global solar UV Index (UVI) as a measure of the amount of solar radiation at the earth's surface [28].

$$\text{UVI} = k_{\text{er}} \int_{250 \text{ nm}}^{400 \text{ nm}} E_{\lambda} S_{\text{er}} \lambda d\lambda \quad (2)$$

According to Eq. (2), UVI is computed as follows: E is the solar spectral irradiance $\text{W}/(\text{m}^2 \text{nm})$ at wavelength, d is the wavelength interval utilized in the summation, S_{er} is the erythema reference action spectrum, and k_{er} is a constant equal to $40 \text{m}^2/\text{W}$. The standard erythema dosage is included in the updated ISO/CIE 17166:2019(en) since the minimum erythema dose is dependent on the individual's susceptibility to UV radiation based on factors such as skin type, time taken to notice erythema, radiometric properties of the source, anatomic location, etc. The UVI values have been divided into exposure groups with designated colors and are displayed in Table 1 to increase public awareness and warn individuals about the erythemal impact of UVR. These figures highlight the amount of UV radiation that can occur throughout the course of a day, and above those values, sun protection measures are required [28].

TABLE 1 UV radiation exposure categories and sun-protective measures.

Exposure category	UVI range	Recommended sun protection
Low	<2	No need for defense. There is no risk of staying outside
Moderate	3–5	Protection is necessary. Midday requires the use of a sunshade
High	6–7	
Very high	8–10	It is necessary to use additional protection such as a hat, shirt, and sunscreen. People should refrain from staying outside, especially around the noon hour

TABLE 2 UPF classifications AS/NZS 4399:2017.

UPF rating	Classification	% of UV radiation blocked
15	Minimum	93.30%
30	Good	96.70%
50, 50+	Excellent	98%

3.1.3 Australian/New Zealand standard AS/NZS 4399-2017

The Australian/New Zealand standard introduced the UPF rating globally. The updated version of this standard, AS/NZS 4399:1996 described the minimum body cover that should be maintained in clothing design. Fabrics can be classified according to the UV-radiation-repellent properties following the index mentioned in Table 2. Other standards involving UV protection properties have been summarized in Table 3 [26,29].

3.2 Antiodor finishing

The major reason for the creation of odor was found to be due to the attack of various bacteria. Hence, the activities of antiodor finishing agents are determined by the methods for determining antimicrobial activities. The main ISO provides three inoculation methods to determine the antibacterial activities of any textile material.

- (a) Absorption method.
- (b) Transfer method.
- (c) Printing method.

In the absorption method, the bacterial suspension used for testing is directly inoculated on the specimen. For the transfer method, the bacteria are transferred into the sample area after placing them on an agar plate. Finally, in the printing method, the bacteria used to print the specimen are kept in a filter.

3.3 Mosquito-repellent finishing

Chemical-based tests can be done in the laboratory with chemicals of known concentration to evaluate the effectiveness of the mosquito-repellent finishes. Two requirements for this testing method are an attractant (human or artificial) and more mosquitoes.

TABLE 3 International standards related to UV protection of textiles.

Name of the standard	Titles	Significance
European Standard EN 13758-1:2002	Standard of analyzing UV-protective properties of apparel or fabric	Applicable to fabric and clothing and different from AS/NZS 4399 in the solar spectrum of Albuquerque
European Standard EN 13758-2:2003	Standard of analyzing UV-protective properties of clothing	Applicable to garments having a UPF rating of 40 or above and a symbol or wording designating UV protection
American Association of Textile Chemists and Colorists Test Method 183-2014	Prevent erythema weighted ultraviolet radiation through fabrics	Measuring technique is similar to EN 13758-1 and UPF is expressed in percentage
UV standard 801	General and unique circumstances	All textile materials that provide UV protection for human skin are applicable
ASTM D 6544-00:2007	Preparing fabrics for UV transmission testing through practice	It is feasible to calculate the fabric's maximum UV transmission during its two-year lifespan.
ASTM D 6603-07:2007	Give appropriate instructions for marking UV protection clothing	Describe UPF value, categorization, and labeling certification in detail

3.3.1 Cage repellency tests

A cage test is a very easy and inexpensive way to determine the mosquito-repellency characteristics of any textile clothing. It is designed for the observation of the mosquitoes' movements around treated and untreated areas. The main advantages of this method are getting information about the biting and at the same time the behavioral differences in mosquitoes. The disadvantage of this method includes a lot of preparation especially taking ethical approval from the participants and also creating real surroundings for the mosquitoes. The mosquitoes used in the test need to be free from any pathogen as humans willingly participate in the test. The dimension of the cage is maintained according to the WHO guideline that suggests a range of 35–40 cm per side for a good result. However, this measurement deviated in some experiments. For example, a cage of 18 × 18 × 18 cm dimension by Bano et al. [17], 30 × 30 × 30 cm by Phasomkusolsil et al. [30], and 35 × 35 × 35 cm cage dimension by Chang et al. [31] were found in various studies. The following stage of the test requires the cage to be concealed with transparent mosquito nets for two purposes, i.e., observing the mosquitoes easily and trapping the mosquitoes inside the cage. There is a hole for arm access that also needs to be covered by the nets properly. The WHO standard suggests that the cage must contain 200 mosquitoes that are starving overnight and that the mosquitoes were given sucrose solution only. An updated version of this standard follows using less number of mosquitoes in the cage (as low as 30 mosquitoes) because less number of mosquitoes is easy to be observed which gives more accurate results to the observer and comfort to the volunteering individual. Volunteers are selected considering that the participants do not smoke usually and they are not allowed to use any kind of fragrance or repellent products for 12 h before the testing. These factors are so important because they can distract the mosquitoes during test time leading to an error in the final results. For preparing the volunteer for the test, their hands are cleaned with unscented soap, rinsed with water, and kept separated by ≥ 20 m away. The arms then are covered with the materials that need to be tested following the insertion of the arms into the cage (Fig. 3A). Both forearms are exposed to the population of the mosquitoes inside the cage for 3 min. If at least two mosquitoes land on the arm within these 3 min, the test continues. On contrary, if no mosquito is found to land in the arm, the arms are withdrawn from the cage. The whole system is observed with a digital camera to get better results. The test is done every 30 min up to 8 h or until the repellency fails. Lupi et al. [32] used a formula to calculate the percentage of repellency or protection time where U indicates the number of mosquitoes on untreated samples and T represents the number of mosquitoes on treated samples. This formula is commonly used and five studies [31,33–36] demonstrated the use of cage tests in their study for mosquito-repellent. The result of protective efficacy (PE) is determined by the following Eq. (3).

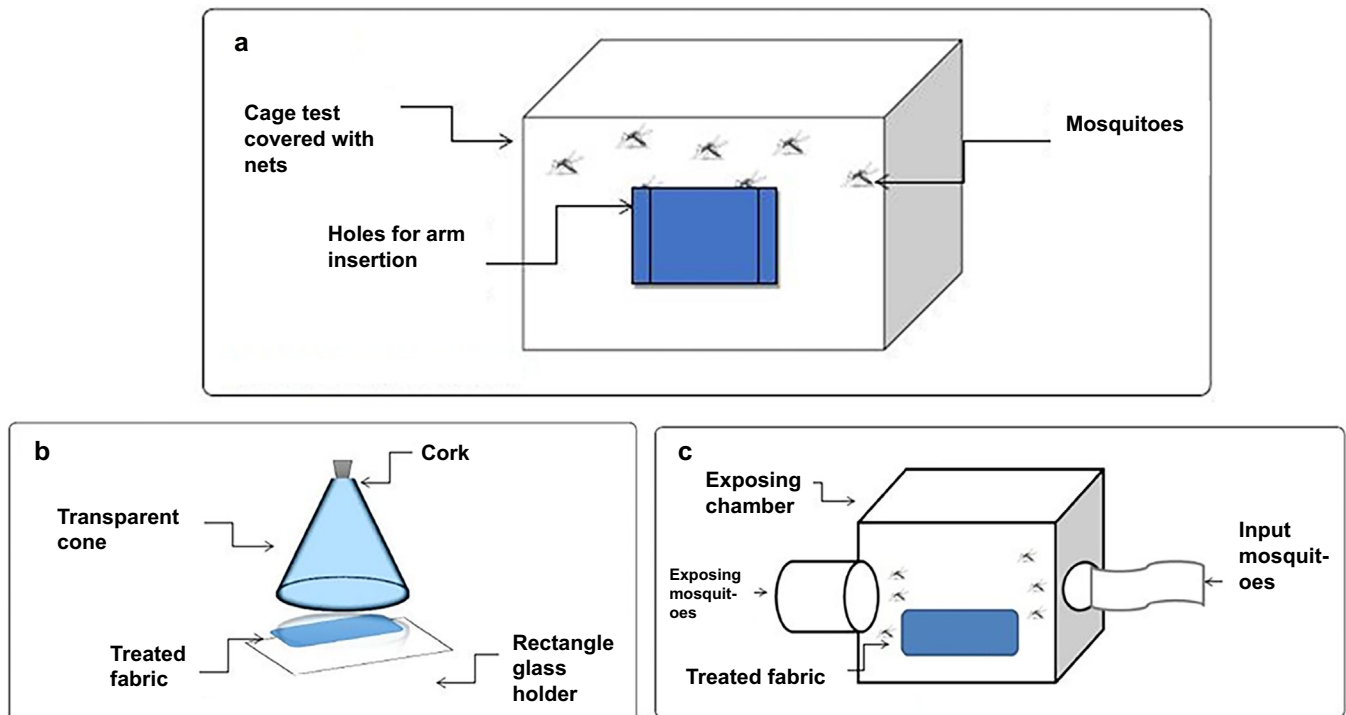


FIG. 3 Illustration of different test procedures for mosquito-repellent; (A) Cage test, (B) Cone test, (C) Excito chamber test.

$$PE = \frac{\text{Mean number of mosquitoes landed on control arm (U)} - \text{mean number of mosquitoes landed on test arm (T)}}{\text{Mean number of mosquitoes landed on the control arm (U)}} \quad (3)$$

3.3.2 Cone test

The cone test can evaluate the resistance of any treated surface against malaria vectors, bioefficacy, and persistence of insecticides following the procedure described by WHO. The advantage of this method is that it does not require any participation of human beings. To evaluate clothes for their ability to repel mosquitoes, researchers do not frequently employ this technique. In the test, fake or animal blood is used as bait to entice mosquitoes looking for hosts. In this test, the treated surface is exposed to the mosquitoes for 3 min. Before that, a cone is applied to the area that has been treated and is taped down (Fig. 3B). Then, 5–10 female mosquitoes were blown into the cone and exposed to the treated surface for 3 min at a temperature of 27°C. This strategy made it simple to observe the behavior of the mosquitoes due to the low mosquito population density. Within 3 min of exposure, the number of mosquitoes sitting on the treated samples was counted. The mosquitoes are then moved from this stage into a plastic cone, where they are maintained under observation while breathing insecticide-free air and receiving a 10% sucrose solution. After an hour, the number of immobilized mosquitoes is counted, and 24 h later, the fatality rate is calculated. The percentage of mosquito-repellency is calculated using the following Eq. (4).

$$\% \text{Mosquito mortality} = \frac{\text{No of Dead specimen}}{\text{Total No. of exposed specimen}} \times 100 \quad (4)$$

3.3.3 Excito chamber test

The excito chamber method has similarities to the cone test where human interactions are not required to perform the test. The chamber is designed with two panels (Fig. 3C), one front and one exit panel with an escape. The mosquitoes are prepared by keeping them starved overnight or at least 4 h before the test. The number of mosquitoes that can escape the chamber and that remain in the chamber is calculated for the eventual calculation. The test is performed for 10–30 min. This procedure is repeated four times and obviously in daylight. The experiment was performed four times in broad daylight. The following Eq. (5) was used to determine the percentage of mosquito-repellency.

$$\% \text{Mosquito mortality} = \frac{(\text{NES} + \text{NDE})}{(\text{NEX})} \times 100 \quad (5)$$

where NES denotes the number of mosquitoes that have escaped, NDE denotes the number of mosquitoes that have died, and NEX denotes the number of mosquitoes that have been exposed.

3.4 Insect-repellent finishing

For assessing the degree of damage to test specimens, two procedures are offered. Only experiments involving carpet beetles employ the excrement weight method. When requested, testing with carpet beetles may also be conducted using the fabric weight loss method in addition to tests with webbing clothing moths. Both of the methods for measuring the insect-repellent finish with the testing procedure are described in the following section.

3.4.1 Excrement weight method

The main principle followed in this method is feeding the test specimen a particular number of insects and measuring the eventual conditions of the insects under specific temperatures and humidity. The main instrument of this method is a cage, which is a large shallow glass or metal flat-bottomed container that facilitates enough space and ventilation so that the specimens can be placed horizontally. The test requires four test specimens and four control specimens to be freed from any dirt and dust. Then the specimens are placed horizontally in the cage followed by placing some larvae on them. Then the cage is covered. The cages containing the test specimens and larvae are kept for 14 days at $27 \pm 1^\circ\text{C}$ ($80 \pm 2^\circ\text{F}$) and $55\% \pm 5\%$ relative humidity without any light being passed.

After the incubation is done, the total alive and dead larvae are counted accurately. The quantity of excrement deposited by larvae on the specimen surface can determine the damage to the fabric. While evaluating the test result, the weight of the excrement and the number of larvae alive should be reported. When an average amount of feces of

not more than 5 mg per specimen is deposited, meaning no single specimen exhibits more than 6 mg of excrement, the examined specimen is deemed to be adequately resistant to carpet beetles. The controls do not exhibit approximately 15 mg per specimen under identical circumstances.

3.4.2 Fabric weight loss method

Two sets of specimens are collected each of them having eight samples; one set contains the test fabric and the others contain the standard control fabric. Four specimens of each set are for the feeding test and the other four of each set are for the humidity test. Four specimens of each set are taken under the same temperature and humidity conditions. Both the humidity control and test specimens are placed in the cage for 48 h before first weighing. Then 10 larvae are placed on the test specimens and the cage is covered for 14 days at $27 \pm 1^\circ\text{C}$ ($80 \pm 2^\circ\text{F}$) and $55\% \pm 5\%$ relative humidity. The loss of weight of the fabric is determined after removing any loose components from the cloth surface, including feces, fraying, cast shells, loose strands, etc. The fabric also should be reconditioned for 24 h before weighing. The loss of weight can be calculated by the following Eq. (6):

$$L = \frac{AC}{B} - D \quad (6)$$

where L is the adjusted calorie restriction from insect feeding expressed in milligrams. A is the four test specimens' combined weights before testing. B is the four specimens used to assess the humidity's weight on average before testing. C is the following testing, the four humidity check specimens' average weight. D is the following testing, the four test specimens' average weight.

3.5 Antioxidant finishing

Ya-Dong Li et al. [37] used ABTS radical cation decolorization assay to determine the activity of the antioxidant finishing. ABTS was mixed with water to a solution of 7 mM concentration, and then the ABTS stock solution was allowed to react with 2.45 mM potassium persulfate to produce the ABTS radical cation (ABTS^+). The mixture was kept to stand in the dark for 12–16 h at room temperature. Before use, the solution was diluted with a phosphate buffer of 0.1 M, and pH 7.4 to reach an absorbance of 0.700 ± 0.025 at 734 nm. In the next stage, 10 mg of fabric sample was immersed into 10 mL of ABTS^+ solution for 30 min. After 30 min, the scavenging capability of ABTS^+ at 734 nm was calculated according to Eq. (7).

$$\text{Antioxidant activity (\%)} = \frac{A_{\text{ctrl}} - A_{\text{spl}}}{A_{\text{ctrl}}} \times 100 \quad (7)$$

where A_{spl} is the absorbance of the residual ABTS^+ in the presence of the fabric sample and A_{ctrl} is the initial absorbance of the ABTS^+ . Three tests for antioxidant activity can be done, and the average value will be reported.

4 Natural dyes and pigments in functional finishing

4.1 UV-protective properties

While there is a wide range of functions to be imparted by synthetic finishing agents, several of these finishes are detrimental to the environment and ecosystem when discharged as effluents. Due to their biodegradable nature, environmentally friendly materials are being used more frequently as a result of growing public awareness of health risks [38]. Several studies on the application of natural dyes as functional finish showed a very good protection against UV radiation. It is quite conspicuous that natural extract from different plants such as *Lawsonia inermis* (Henna leaves), *Acacia nilotica* (Kikar bark), *Eucalyptus globules* (Tasmanian blue gum), *Punica grantum* (Pomegranate peel), *Allium cepa* (Onion peel), and *Tagetes erecta* (Marigold flowers) show best protection against UV radiation [39]. The main reason behind the protection of these natural sources from sun radiation is containing some bioactive constituents such as gallic acid, ellagic acid, carotenes, anthocyanins, and tannins.

Henna with the scientific name, *Lawsonia inermis* and another alternative name, lawsone contains 2-hydroxy-1,4-naphthoquinone, also called hennotannic acid in its chemical structure and physical appearance is illustrated in Fig. 4A. Because of this chemical component, the extract of henna leaves provides red-orange color on the substrate. According to the CI classification, the produced color corresponds to Natural Orange 6 with CI number 75480 [40]. The combination of active components in the henna leaves is lawsone, gallic acid, xanthenes, white resin, carbohydrates,

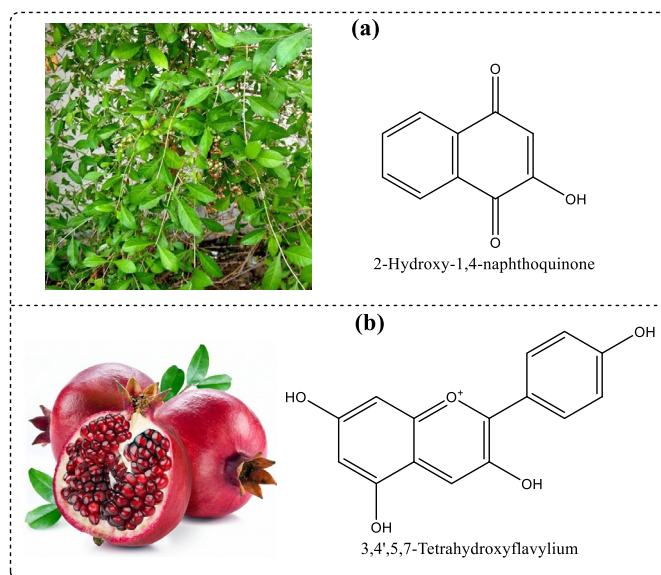


FIG. 4 Physical appearance and chemical structure of (A) Henna (*Lawsonia*) and (B) *P. grantum*. Tannins.

and tannin. Besides these coloration properties, henna is a very good source of UV protection for cotton, linen, and wool fabric.

Linen is one of the vegetable fibers that can be modified through mordanting and dyeing processes when processed in fabric form. The improvement of fabric in UV protection is increased with the concentration of dye and the mordant. The sample displayed a UPF value of 45, which is regarded as providing effective protection. Henna demonstrated the highest absorption in the UV area, as was stated in the UV–visible study, which may be a contributing factor to the UV protection it imparts. The UPF values showed a decreasing trend after 20 wash cycles and it reached 29.20 which indicates the moderate performance of UV [41].

Dyes extracted from the leaves of the eucalyptus create a yellowish-brown effect on the fabric surface. From the functional perspective, the dyeing can protect the fabric from UV radiation when treated with mordants. The main components of eucalyptus are quercetin, tannin (gallic acid and ellagic acid), and flavonoids (quercetin, rutin, etc.) that help in coloration and fixation of the color. The application of the mordant especially FeSO_4 causes the treated fabric to show protection against UV radiation [42]. Another study worked for dyeing wool fabric with natural dyes, extracted from henna that prevents the fabric from UV rays. The FTIR analysis of the dyed sample provided information about the presence of a hydroxyl group and two carbonyl groups that eventually indicates the presence of henna. The sample later was tested and it showed a very good fastness towards the light, i.e., UV ray.

Cotton fabric dyed with the extract of the *P. grantum* showed good UV protection performance. Pomegranate with the scientific name *P. grantum* is generally known as a fruit item with the main chemical component of Tannins. The physical appearance and chemical structure of *P. grantum* is shown in Fig. 4B. The extract from pomegranate provides brown, yellow, and black shades on the fabric. Besides being popular in the field of natural dye, the component tannin has a very good tendency to provide UV protection from sun radiation. On comparison with other natural sources, it was found that this component can yield maximum protection against UV radiation. The mean UPF value of cotton fabric came to 71.6 which is treated by *P. grantum* ensuring an excellent level of UV protection [39]. The details of natural dyes used as a UV-protective finish with the performances are illustrated in Table 4.

4.2 Antiodor properties

Among the growing interest in imparting multifunctional effects on fabric or textile materials, antiodor properties are becoming very common. Textile materials including clothing and other form are very susceptible to microorganisms, bacteria, and molds. Cotton fibers have an intrinsic property to allow microbial growth in clothing. In addition, synthetic fiber-based clothing and other materials also require more attention for antiodor as they have limitations of moisture transportation capability and thus contain perspiration that may create wetness and microbial components in them [45]. This phenomenon is particularly common in sportswear causing unpleasant odors.

TABLE 4 Details of using natural dyes as UV-protective finish.

Serial no.	Types of fabric	Natural dyes or pigments	Treated conditions (such as M:L, pH, temperature, time, mordant, etc.)	Performances			References
				UV-A	UV-B	UPF	
1	Cotton	Henna	M:L-1:25, Temp: 70°C, Time: 100 min	87	88	38	[39]
2	Cotton	Pomegranate peel	M:L-1:15, Temp: 60°C, Time: 60 min	88	89	71.6	[39]
3	Linen	Henna	M:L-1:20, Temp: 90°C, Time: 1 h, Mordant: Copper sulfate			54.1	[41]
4	Cotton	Green tea	M:L-1:30, Temp: 105°C, Time: 105 min	91.3	92.8	–	[43]
5	Cotton and silk	Rheum root and <i>Lithospermum erythrorhizon</i> root	M:L-1:40, Temp: 100°C, Time: 60 min	97.5	97.5		[44]

TABLE 5 Details of using natural dyes as antiodor finish.

Sl. no.	Types of fabric	Natural dyes or pigments	Treated conditions (such as M:L, pH, temperature, time, mordant, etc.)	Performances	References
1	Cotton	Onion	M:L-1:20, Temp: 85°C, Time: 1 h	Excellent	[46]
2	Modal	Burberry	M:L-1:20, Temp: 70°C, Time: 2 h	Very good	[40]

Natural dyes from pomegranate, annatto, myrobalan, onion, indigo, and barberry with scientific names *P. granatum*, *Bixa orellana*, *Terminalia chebula*, *A. cepa*, *Indigofer atinctoria*, and *Berberis vulgaris*, respectively, can be used to extract dye and eventually used for antiodor finish. If the cotton fabrics are treated with sodium hydroxide and an aqueous solution of morpholine cellulase enzyme with different concentrations, time, and temperature, and eventually dyed with the dye extracted from onion, the fabric shows antiodor properties [40,46]. The subjective evaluation of the dyed fabric was conducted on six female participants with an age range of 25–50 years and the result was rated between 0 (repulsive) and 10 (ideal). The overall antiodor evaluation was in the range of good to excellent with a value of 7–9. The details of natural dyes used as an antiodor finish with the performances are represented in Table 5.

4.3 Mosquito-repellent properties

Many illnesses, including malaria, lymphatic filariasis, western equine encephalitis, etc., are spread by female *Culex quinquefasciatus* mosquitoes, which are regarded to be house mosquitoes. Mosquitoes were chosen for laboratory testing because their larvae develop into pupae consuming sugar and finally adult mosquitoes in 36 h at temperatures between 27°C and 30°C [47]. The chemically active components found in natural colors include tannins, polyphenols, phenols, carotenes, quercetin, anthocyanins, etc. These chemicals are capable of infiltrating and interfering with all biological processes in a variety of insects, including moths, carpet beetles, mosquitoes, and others. They play a critical role in interrupting and spreading the life cycles of these insects and protecting people from their diseases [48].

The clothes colored with *T. erecta* extract had the highest degree of mosquito repulsion (66.25%), which is consistent with previously published findings [49]. *T. erecta* contains various secondary metabolites with antifeedant activities [50]. Major antifeedant activities were principally found in three terpenoid groups: neoclerodanediterpenes, neoclerodanesesquiterpenes, and limonoids [50]. When given to insects, antifeedant chemicals temporarily or permanently suppress the insects' appetites. Terpenoid, an active compound found in *T. erecta* leaves and flowers has been widely documented to have antimosquito and biocidal properties. The primary bioactive components in *T. erecta* with biocidal and repulsive properties have been hypothesized to be its terpenoid content [51]. The details of natural dyes and pigments used as a mosquito-repellent finish with the performances are shown in Table 6.

4.4 Insect-repellent properties

Textile applications including clothing, technical textiles, and home appliances that are warm, soft, and flame retardant, are fabricated by protein fibers such as wool and silk and are very prone to insect attack. Cloth moth (*Tineola bisselliella*) and carpet beetle (*Anthrenus verbasci*) are common insects including their larvae that attack wool materials.

TABLE 6 Details of using natural dyes as mosquito-repellent finish.

Sl. no.	Types of fabric	Natural dyes or pigments	Treated conditions (such as M:L, pH, temperature, time, mordant, etc.)	Performances	References
1	Cotton	<i>T. erecta</i> (Marigold flower)	M:L-1:12, Temp: 100°C, Time: 60 min	Highest performance 66.25%	[49]
2	Cotton	Henna	M:L-1:5, Temp: 70°C, Time: 60 min	Moderate performance 48%	[39]

TABLE 7 Details of using natural dyes as insect-repellent finish.

Sl. no.	Types of fabric	Natural dyes or pigments	Treated conditions (such as M:L, pH, temperature, time, mordant, etc.)	Performances	References
01	Wool	Anthraquinone dyes	M:L = 40:1 pH = 5.5 Temperature = $93 \pm 2^\circ\text{C}$ Time = 45 min Mordant = aluminum sulfate, potassium dichromate, copper sulfate, iron oxide, and stannous chloride	The sample showed the least amount of damage which indicates the good insect-repellency of anthraquinone dyes	[52]
02	Wool	Flavonoid dyes	M:L = 40:1 pH = 5.5 Temperature = $93 \pm 2^\circ\text{C}$ Time = 45 min Mordant = aluminum sulfate, potassium dichromate, copper sulfate, iron oxide, and stannous chloride	The sample showed a good amount of weight loss that indicates the least insect-repellency of flavonoid dyes	[52]

Some chemical/synthetic finishing agents such as DDT (dichlorodiphenyltrichloroethane), permethrin, permethrin/hexahydro pyrimidine derivative, cyhalothrin, etc., are found effective to be used as insect-repellent materials. Nanoparticles of TiO_2 were used as a repellent agent for a carpet beetle to protect the wool-based textile materials. However, due to the creation of disturbance in ecology, chemical-based antiinsect agents are not welcome, and alternative natural sources have been investigated. Park et al. [52] studied eight types of natural dyes for evaluating the insect-repellent properties of the dyes where anthraquinone dyes from cochineal, madder, and walnut, and flavonoid dyes from chestnut, fustic, indigo, and logwood were applied on wool fabric. Five types of mordants were used to perform the dyeing procedure and AATCC Test Method 24-1999 was used to evaluate the performance of the insect-repellency. The details of the procedure and performance results are described in Table 7.

4.4.1 Anthraquinone dyes

Anthraquinone dyes, the natural dyes, are used as the main coloring agents and are found in red color and are also called mordant dyes. Madder, lacs, kermes, and cochineal natural dyes contain anthraquinone chemical structures (Fig. 5A) [53]. These dyes can improve the UV protection performance, insect-repellent performance, and antibacterial activity (against *Staphylococcus aureus* (*S. aureus*) and *Escherichia coli* (*E. coli*)) of the natural fabric).

4.4.2 Flavonoid dyes

The flavonoid class dyes contain two benzene rings linked through a heterocyclic pyrene ring (Fig. 5B). Sources of some of these dyes include *A. cepa* (onion), *Artocarpus heterophyllus*/*Artocarpus integrifolia* (jackfruit), *Myrica esculenta* (kaiphal), *Cannabis sativa* (hemp), *Delphinium zaili* (yellow larkspur), *Gossypium herbaceum*, *Sophora japonica*/*Styphnolobium japonicum*, and others are other plant sources. *Bignonia chica* (Carajuru), *Butea monosperma*/*Butea frondosa* (flame of the forest/palas), *Mallotus philippinensis* (Kamala), and *Pterocarpus santalinus* (red sandalwood) [54,55].

4.5 Antioxidant properties

The concern about the environment has led researchers to think about the use of natural dye. At the same time, the value addition to textile materials in terms of different functionalities is very common nowadays. The antioxidant

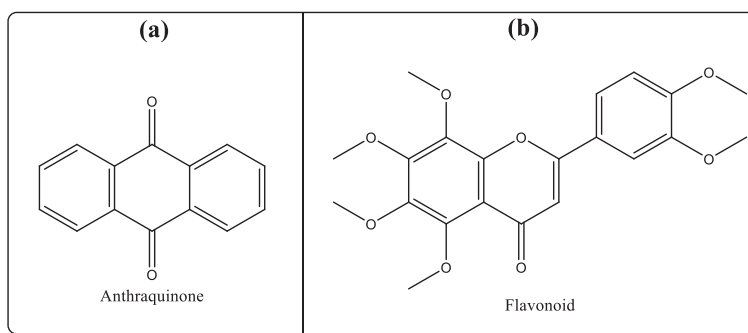


FIG. 5 Chemical structure of (A) Anthraquinone and (B) Flavonoid.

property of any textile material is also being incorporated to a great extent. The use of natural dyes as a multifunctional component in textile materials is described in this section with the details of the sources. The advantages of natural dyes are that the dyes contain chemical components that work both for the coloring of the fabric and simultaneously other functional agents. Some sources of natural dyes are described in the following section with the name of their chemical contents that have dual or multiple compatibility for dyeing and finishing of the clothes.

A. nilotica, used mainly as a coloring agent, is a medicinal plant of the family *Fabaceae* and contains high content of polyphenols and flavonoids with quercetin, acacetin, and ellagitannins. The main coloring elements of *A. nilotica* are quercetin, acacetin, and ellagitannins (Fig. 6A). Apart from the coloring ability, *A. nilotica* has intrinsic properties of fluorescence emission [56] and antioxidants [57]. The powder form of this plant is dark brown.

T. arjuna contains a high level of polyphenols and is a known Ayurvedic drug. This plant is indigenous to the Indian peninsula and also is abundantly found in the sub-Himalayan tract. The main coloring components of this plant are baicalein and ellagic acid (Fig. 6B). Besides coloring, the extract of this plant was found effective in antioxidant properties [57].

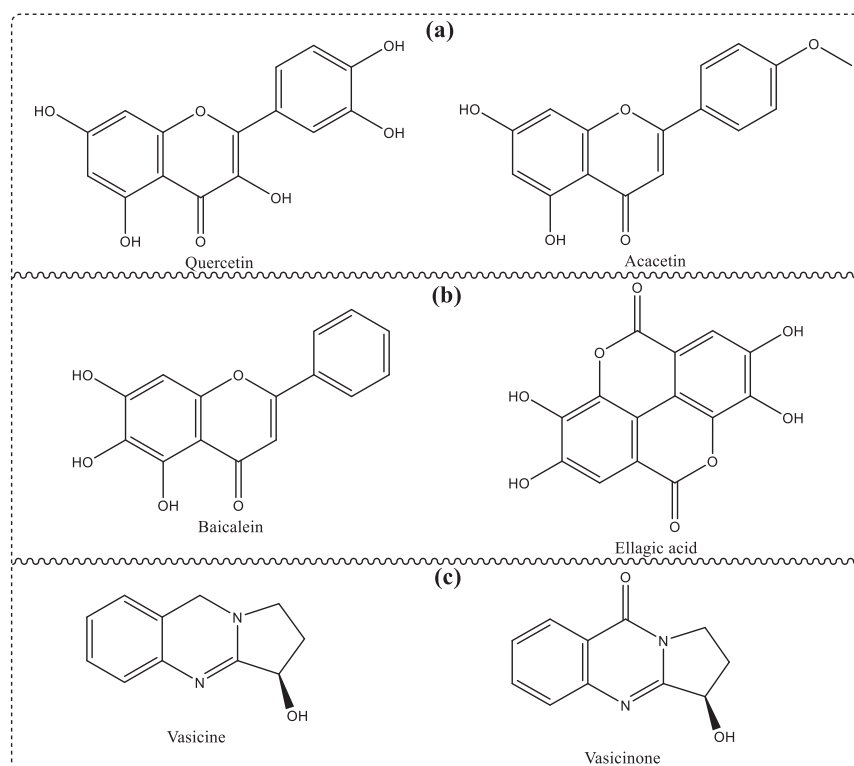


FIG. 6 Illustration of the chemical components in (A) *A. nilotica*, (B) *T. arjuna*, and (C) *A. vasica*.

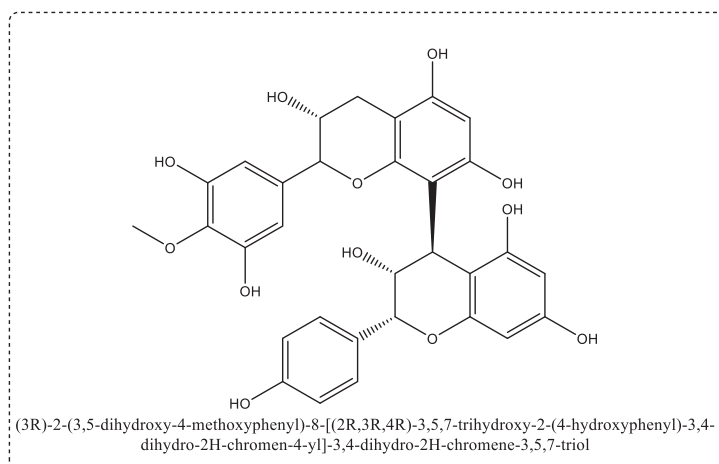


FIG. 7 Illustration of the chemical structure of proanthocyanidin.

A. vasica which belongs to the family *Acanthaceae* is also known for Ayurvedic and Unani medicine. This plant is available in the Indian peninsula up to an altitude of 1300 m. The main chemical compounds in *A. vasica* are vasicine and vasicinone (Fig. 6C).

The grape has many bioactive phenol compounds that are very useful in some biological activities such as antioxidant, cardioprotective, anticancer, antiinflammation, antiaging, and antimicrobial properties. Among the whole grape, the main component of its seed is proanthocyanidin (condensed tannins) [58] which consists of a series of polymerized flavan-3-ols linking through the 4 and the 8 positions, as shown in Fig. 7. The extraction of seed in powdered form can be used as both dyestuff and antioxidant agents [51]. The details of natural dyes and pigments used as an antioxidant finish with the performances are shown in Table 8.

TABLE 8 Details of using natural dyes as antioxidant finish.

Sl. no.	Types of fabric	Natural dyes or pigments	Treated conditions					Performances	References
			M:L	pH	Temperature (°C)	Time (min)	Mordant		
01	Cotton	Extract of <i>Araucaria columnaris</i>	1:40	5	90	90	Myrobalan, alum	Strong antioxidant activity	[37]
02	Wool	Extract of <i>A. nilotica</i> dye	1:50	3	90	60	Alum, ferrous sulfate, stannous chloride	Mordanting reduced the antioxidant potential of dyed woolen yarn	[59]
03	Wool	Cochineal and Madder dyes	1:50	–	90	–	–	Enhanced antioxidant activity	[60]
04	Cotton	Extract of Grape seed extract (GSE)	1:50	4–8	60–90	30–90	–	20%owf GSE had a high antioxidant activity of 97%	[61]
05	Wool	<i>A. nilotica</i> , <i>T. arjuna</i> , and <i>A. vasica</i>	1:50	7	90	45	Ferrous sulfate	Antioxidant properties were developed	[57]
06	Wool	Marigold flowers dyestuff	1:40	7	90	60	Alum, iron sulfate, tin chloride	Antioxidant activity up to more than 95%	[62]

5 Conclusion

Natural dyes and pigments not only create soft and muted tones, but they also can develop innovative features that might be used to achieve an active textile substrate with performance capabilities such as deodorizing, antioxidant, antibacterial, antifeedant, UV protection, mosquito repellent, and other similar attributes. In the current market, the production of colored textile items that use natural colorants has the potential to increase customer interest to an even higher degree. As a result, one of the most prominent current developments in this industry is the search for natural dyes and pigments that can be degraded naturally. To this point, a variety of natural plants have shown encouraging results in the functional finishing of textiles, enhancing both the industry's sustainability and its friendliness to the environment. There is no doubt in the widespread recognition that natural dyes and pigments can be used as potential alternatives for synthetic chemicals in some categories of textile finishing purposes. This helps to minimize the harmful effects that synthetic dyes have on both human health and the environment. This chapter mainly focused on the functional finishing of textile materials using various types of natural dyes and pigments. The different sources of natural dyes used for functional finishing such as UV-protective, mosquito-repellent, antiodor, insect-repellent, and antioxidant are described by their chemical structures. The testing methods for functional finishing are also discussed in this chapter. Therefore, this chapter provides a complete outline of functional finishing using natural dyes and pigment.

5.1 Future perspectives

The production of textiles in the future must take into account the concept of sustainability, a goal that may now also be employed for marketing purposes. In point of fact, over the past several years, a multitude of businesses, both large and little, have made efforts to introduce environmentally friendly materials onto the market. The public has shown a range of reactions regarding their interest in this topic, and in the end, the global market remained unchanged. In addition, the utilization of natural dyes and pigments may appear to be a sustainable option. Despite this, we should back the concept of recycling old materials over and again. In the not-too-distant future, the subject of safety considerations during the treatment of textiles will be a matter of concern. Taking into account the fact that natural dyes need a significant amount of raw material, this can result in better use of natural coloring agents as well as functional finishing. During the washing process, synthetic textiles are known to release hazardous ingredients into the environment. By the way, natural dyes and pigments show less durability than synthetic chemicals. Therefore, future research can be more focused on increasing the finishing durability of textile materials using natural dyes and pigments.

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